

A combinatorial proof of the M-convexity of cylindric skew Schur polynomials

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Abstract

Wang, Zhang and Zhang [WZZ, arXiv:2401.14632, Problem 5.2] ask for a combinatorial proof of the M-convexity of cylindric skew Schur polynomials *based on their tableau interpretation*. Their own Corollary 4.9 obtains M-convexity, but only by importing two external results — Lam’s Corollary 8.5 (affine Stanley symmetric functions expand positively in dual k -Schur functions) and Lee’s Corollary 5 — and the sibling Problem 5.1 makes it explicit that what is wanted is a proof that does *not* route through Lam’s theorem.

We give such a proof. Working in the periodic Maya-diagram (bead) model, a semistandard cylindric skew tableau becomes a chain of n -periodic bead configurations, and its weight becomes the increment sequence of the single scalar statistic $u^t = \sum_{i=1}^m x_i^t$. The entire argument then rests on one observation (Lemma 3.1): *with its two neighbours held fixed, a row of the chain varies independently in each bead, over a product of integer intervals $\prod_i [L_i, U_i]$, whose endpoints satisfy the identity $\sum_i (L_i + U_i) = \sum_i (p_i + r_i)$* (Lemma 3.2). From this we obtain, in turn, a cylindric Bender–Knuth involution, the exchange move (a single bead sliding one site), the dominance-ideal property of the weight set, and — via a greedy chain — the unique dominance maximum. A self-contained polymatroid argument converts this into M-convexity.

The proof consumes no external theorem: not Lam’s Corollary 8.5, not Lee’s Corollary 5, not Rado’s theorem, not even Postnikov’s symmetry theorem for cylindric skew Schur functions (which we reprove along the way, Corollary 3.4). It also produces more than M-convexity: it identifies the Newton polytope explicitly as the $\hat{\lambda}$ -permutahedron for a partition $\hat{\lambda}$ computed by an explicit greedy algorithm.

1 Introduction

1.1 The problem

Let $f = \sum_{\alpha} c_{\alpha} x^{\alpha} \in \mathbb{R}[x_1, \dots, x_{\ell}]$ and let $\text{supp}(f) = \{\alpha : c_{\alpha} \neq 0\}$. A set $J \subseteq \mathbb{N}^{\ell}$ is *M-convex* if for all $\alpha, \beta \in J$ and every index i with $\alpha_i > \beta_i$ there is an index j with $\alpha_j < \beta_j$ and $\alpha - e_i + e_j \in J$. A polynomial is M-convex if its support is.

Cylindric skew Schur functions were introduced by Postnikov. Wang–Zhang–Zhang (henceforth WZZ) prove in [1, Cor. 4.9] that every cylindric skew Schur polynomial is M-convex. Their route is: M-convexity of dual k -Schur polynomials (their Theorem 3.6, a genuinely combinatorial theorem about k -Kostka numbers) $\Rightarrow$ M-convexity of affine Stanley symmetric polynomials (their Theorem 4.4, which invokes [2, Cor. 8.5]) $\Rightarrow$ M-convexity of cylindric skew Schur polynomials (their Cor. 4.9, which invokes [3, Cor. 5]).

They then pose, verbatim:

Problem 1.1 (WZZ, Problem 5.2). Find a combinatorial proof of the M-convexity of cylindric skew Schur polynomials based on their tableau interpretation.

The motive is stated one paragraph earlier, attached to the sibling Problem 5.1: “*One of the key ingredients of our approach is Theorem [Lam, Cor. 8.5], a deep result obtained by Lam. It would be interesting to find a direct proof of Theorem [Thm. 4.4] based on the definition given by (4.1).*” Problem 5.2 is posed as “similar to Problem 5.1” and inherits this motive. So the object of the exercise is a proof that bypasses the import chain. Since the truth of the statement is not in question, the success criterion is checkable: *if the argument still consumes Lam’s Corollary 8.5 or Lee’s Corollary 5, it has not answered the problem.* We therefore state explicitly, in §8, every external result the argument uses. The list is empty.

1.2 What we prove

Theorem 1.2 (Main theorem; see Theorem 7.1). *Let λ/μ be a cylindric skew shape in $\mathcal{C}^{n,m}$ and let $\ell \geq 1$. If the cylindric skew Schur polynomial $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ is nonzero, then there is a partition $\hat{\lambda} = \hat{\lambda}(\lambda/\mu, \ell)$ of $|\lambda/\mu|$, computed by an explicit greedy algorithm, such that*

$$\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell.$$

In particular $s_{\lambda/\mu}^c$ is M-convex, is SNP, and $\text{Newton}(s_{\lambda/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$ is a $\hat{\lambda}$ -permutahedron.

This is strictly stronger than WZZ’s Corollary 4.9: it names the polytope.

1.3 The shape of the argument, and why it is short

The tableau interpretation of $s_{\lambda/\mu}^c$ is a chain $\mu = \lambda^0 \subseteq \lambda^1 \subseteq \dots \subseteq \lambda^\ell = \lambda$ of cylindric shapes with cylindric-horizontal-strip steps, weighted by $\alpha_t = |\lambda^t/\lambda^{t-1}|$. In the periodic Maya-diagram model a cylindric shape becomes a strictly increasing $x : \mathbb{Z} \rightarrow \mathbb{Z}$ with $x_{i+m} = x_i + n$ (§2), and two things happen at once:

- (i) **The weight linearises.** Putting $u^t = \sum_{i=1}^m x_i^t$, one has $\alpha_t = u^t - u^{t-1}$. The weight of a cylindric tableau is the increment sequence of a single integer statistic. Consequently $\alpha_1 + \dots + \alpha_r = u^r - u^0$ telescopes, which is what makes the greedy argument of §5 work.
- (ii) **The chain decouples.** The condition that $\lambda^{t-1} \subseteq \lambda^t$ and $\lambda^t \subseteq \lambda^{t+1}$ are both horizontal strips involves x_i^t only through x_i^{t-1} and x_i^{t+1} — never through x_j^t for $j \neq i$. So with its neighbours fixed, the row x^t ranges over a *box* $\prod_i [L_i, U_i]$ (Lemma 3.1).

Everything else is bookkeeping on that box. The one identity that has to be checked is

$$\sum_{i=1}^m (L_i + U_i) = \sum_{i=1}^m (p_i + r_i), \quad p = x^{t-1}, \quad r = x^{t+1},$$

and its proof (Lemma 3.2) is “max + min = sum” plus one wrap-around term. It is worth flagging that the wrap-around contributes exactly $+n$, and that without it the identity would read $\sum (L_i + U_i) = \sum (p_i + r_i) - n$. *The cylindric case works because of the periodicity, not in spite of it.* (I record this because I first derived the wrong version, by dropping the wrap-around term; the computation in §9 caught it.)

1.4 Relation to the ribbon/Maya move of Naprienko

Naprienko [4, Lemma 4.4] records that if λ/μ is a *ribbon* with minimal content k and maximal content K , then $\alpha_0(\lambda) = (\alpha_0(\mu) \setminus \{k\}) \cup \{K+1\}$ — one bead hops from site k to site $K+1$, possibly over other beads. That is a natural candidate for an “exchange move”. It is not the move used here, and the distinction matters:

- a **ribbon** of size r is one bead hopping r sites, jumping freely over occupied sites;
- a **horizontal strip** of size r is several beads hopping right with pairwise disjoint closed trajectories — no bead may pass or land on another (Lemma 2.4).

Cylindric skew Schur functions are built from horizontal strips, so the ribbon move is the wrong local move for this problem. The move that actually implements the M-convex exchange axiom is smaller still: *one bead in one row of the chain slides one site to the left* (Corollary 3.5).

2 The bead model

Fix integers $n > m \geq 1$.

Definition 2.1. A *cylindric shape of type (n, m)* is a strictly increasing map $x : \mathbb{Z} \rightarrow \mathbb{Z}$, written $i \mapsto x_i$, satisfying $x_{i+m} = x_i + n$ for all $i \in \mathbb{Z}$. We write $\mathcal{C}^{n,m}$ for the set of these. A shape is determined by the tuple $(x_1, \dots, x_m)$, which satisfies $x_1 < x_2 < \dots < x_m < x_1 + n$. Set

$$u(x) = \sum_{i=1}^m x_i.$$

For $\nu, \rho \in \mathcal{C}^{n,m}$ write $\nu \subseteq \rho$ if $\nu_i \leq \rho_i$ for all i , and in that case put $|\rho/\nu| = u(\rho) - u(\nu)$. Write $\nu \prec \rho$ (“ ρ/ν is a horizontal strip”) if

$$\nu_i \leq \rho_i < \nu_{i+1} \quad \text{for all } i \in \mathbb{Z}. \quad (1)$$

By periodicity it suffices to check (1) for $i = 1, \dots, m$, reading $\nu_{m+1} = \nu_1 + n$.

2.1 Dictionary to the lattice-path definition

We recall the definition used by WZZ [1, §5], following Lam and Postnikov. A cylindric shape is an infinite lattice path in $\mathbb{Z}^2$ with steps upwards and to the left, invariant under the shift $(m-n, m)$, where $1 \leq m \leq n-1$; $\mu \subseteq \lambda$ means μ lies weakly to the left of λ ; and a *semistandard cylindric skew tableau* of shape λ/μ and weight $\alpha = (\alpha_1, \dots, \alpha_\ell)$ is a chain $\mu = \lambda^0 \subseteq \lambda^1 \subseteq \dots \subseteq \lambda^\ell = \lambda$ of cylindric shapes such that each λ^t/λ^{t-1} contains at most one box in each column and α_t boxes in any $n-m$ consecutive columns. Then $s_{\lambda/\mu}^c = \sum_T x^{\text{weight}(T)}$.

Proposition 2.2. *Index the steps of such a path P by $t = y - x$, where (x, y) is the starting point of the step (this is well defined: every up step and every left step increases $y - x$ by exactly 1). Let $A(P) \subseteq \mathbb{Z}$ be the set of indices of the up steps, and let $x_1 < x_2 < \dots$ be its elements in increasing order, indexed so that $x_{i+m} = x_i + n$. Then $P \mapsto (x_i)$ is a bijection from cylindric shapes onto $\mathcal{C}^{n,m}$, and under it:*

1. $\mu \subseteq \lambda$ (weakly to the left) $\iff \mu_i \leq \lambda_i$ for all i ;

2. the number of boxes of λ/μ in any $n - m$ consecutive columns equals $u(\lambda) - u(\mu)$;
3. λ/μ has at most one box in each column $\iff \mu \prec\prec \lambda$.

Proof. That $t = y - x$ is a well-defined step index: an up step sends $(x, y) \mapsto (x, y + 1)$ and a left step sends $(x, y) \mapsto (x - 1, y)$, so in both cases $y - x$ increases by 1; thus the steps of P are indexed by $\mathbb{Z}$ with step t starting at the unique point of P on the line $y - x = t$. The shift $(m - n, m)$ sends $y - x \mapsto y - x + n$, so $A(P) + n = A(P)$; it carries m up steps to m up steps per period, so $|A(P) \cap [t, t + n)| = m$. Conversely any n -periodic subset with m elements per period is $A(P)$ for a unique such path (reconstruct P by following the prescribed steps from the point of P on $y - x = 0$, which is determined by $A(P)$ up to the shift already accounted for). This is the bijection.

For (i)–(iii) it is enough to verify the statements in a bounded window of rows, because all three conditions are local (they involve finitely many rows and columns at a time) and both sides are invariant under the shift. In a bounded window a cylindric shape is the boundary path of an ordinary partition, $A(P)$ restricted to the window is its β -set $\{\lambda_i - i\}$, and (i)–(iii) become the classical statements $\mu \subseteq \lambda \iff \mu_i \leq \lambda_i$; $|\lambda/\mu| = \sum_i (\lambda_i - \mu_i) = \sum_i ((\lambda_i - i) - (\mu_i - i))$; and λ/μ a horizontal strip $\iff \lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \dots \iff$ the β -sets interlace as in (1). $\square$

Remark 2.3. Proposition 2.2 is the only point at which conventions could go wrong, and it is the point at which an error would be invisible — one would prove a correct theorem about the wrong object. It is therefore checked twice computationally in §9: once against an independently coded ordinary skew Kostka computation (the “unrolled” limit $n \gg 0$), and once against WZZ’s own published Example 4.2, which the model reproduces exactly including its negative Schur coefficient.

For the record we also note the bead description of a horizontal strip, since it is what distinguishes the present move from the ribbon move of [4].

Lemma 2.4. $\nu \prec\prec \rho$ if and only if the closed intervals $[\nu_i, \rho_i]$, $i \in \mathbb{Z}$, are pairwise disjoint. Equivalently: the beads move to the right, and no bead’s trajectory contains another bead’s initial or final position.

Proof. $[\nu_i, \rho_i]$ and $[\nu_{i+1}, \rho_{i+1}]$ are disjoint iff $\rho_i < \nu_{i+1}$, since $\nu_i < \nu_{i+1}$ and $\rho_i < \rho_{i+1}$; and the intervals are increasing in i , so pairwise disjointness reduces to consecutive disjointness. $\square$

2.2 Tableaux as chains of bead configurations

By Proposition 2.2, a semistandard cylindric skew tableau of shape λ/μ with ℓ letters is exactly a sequence $x^0, x^1, \dots, x^\ell \in \mathcal{C}^{n,m}$ with $x^0 = \mu$, $x^\ell = \lambda$, and $x^{t-1} \prec\prec x^t$ for $1 \leq t \leq \ell$; its weight is

$$\alpha_t = u(x^t) - u(x^{t-1}), \quad 1 \leq t \leq \ell. \quad (2)$$

Write $\mathcal{T}_\ell(\lambda/\mu)$ for the set of such chains and

$$W_\ell(\lambda/\mu) = \{\alpha(T) : T \in \mathcal{T}_\ell(\lambda/\mu)\} = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) \subseteq \mathbb{N}^\ell.$$

Every element of W_ℓ has coordinate sum $d := |\lambda/\mu| = u(\lambda) - u(\mu)$.

3 The decoupling lemma

Throughout this section $p, r \in \mathcal{C}^{n,m}$ are fixed. Define, for $i \in \mathbb{Z}$,

$$L_i = \max(p_i, r_{i-1} + 1), \quad U_i = \min(p_{i+1} - 1, r_i). \quad (3)$$

Both are n -periodic in the sense $L_{i+m} = L_i + n$, $U_{i+m} = U_i + n$.

Lemma 3.1 (Decoupling). $\{x \in \mathcal{C}^{n,m} : p \ll x \text{ and } x \ll r\} = \{x : \mathbb{Z} \rightarrow \mathbb{Z} \text{ with } x_{i+m} = x_i + n \text{ and } L_i \leq x_i \leq U_i \forall i\}$. In particular this set is a product of integer intervals, and it is nonempty if and only if $L_i \leq U_i$ for all i .

Proof. $p \ll x$ reads $p_i \leq x_i < p_{i+1}$, i.e. $p_i \leq x_i \leq p_{i+1} - 1$. And $x \ll r$ reads $x_i \leq r_i < x_{i+1}$; the second half, reindexed, is $x_i > r_{i-1}$, i.e. $x_i \geq r_{i-1} + 1$. Together these are exactly $L_i \leq x_i \leq U_i$, and they constrain x_i only through p and r — never through x_j for $j \neq i$.

It remains to check that any n -periodic x with $L_i \leq x_i \leq U_i$ is automatically strictly increasing, hence a cylindric shape. Indeed $x_i \leq U_i \leq p_{i+1} - 1 < p_{i+1} \leq L_{i+1} \leq x_{i+1}$. $\square$

Lemma 3.2 (The palindrome identity). Suppose $L_i \leq U_i$ for all i . Then

$$\sum_{i=1}^m (L_i + U_i) = \sum_{i=1}^m (p_i + r_i).$$

Proof. Reindex the U -sum by $j = i + 1$:

$$\sum_{i=1}^m U_i = \sum_{j=2}^{m+1} \min(p_j - 1, r_{j-1}).$$

The $j = m + 1$ term is $\min(p_{m+1} - 1, r_m) = \min(p_1 + n - 1, r_0 + n) = \min(p_1 - 1, r_0) + n$, which is the $j = 1$ term plus n . Hence

$$\sum_{i=1}^m U_i = \sum_{j=1}^m \min(p_j - 1, r_{j-1}) + n = \sum_{j=1}^m (\min(p_j, r_{j-1} + 1) - 1) + n.$$

Since $\max(a, b) + \min(a, b) = a + b$,

$$\sum_{j=1}^m (L_j + U_j) = \sum_{j=1}^m (p_j + r_{j-1} + 1) - m + n = \sum_{j=1}^m p_j + \sum_{j=1}^m r_{j-1} + n.$$

Finally $\sum_{j=1}^m r_{j-1} = \sum_{j=0}^{m-1} r_j = \sum_{j=1}^m r_j - r_m + r_0 = u(r) - n$, and the claim follows. $\square$

Remark 3.3. The term $+n$ produced by the wrap-around is exactly what makes the identity hold; a computation that forgets it yields $\sum (L_i + U_i) = u(p) + u(r) - n$, and with it the false conclusion that the cylindric analogue of Bender–Knuth fails.

Now fix a chain $T = (x^0, \dots, x^\ell) \in \mathcal{T}_\ell(\lambda/\mu)$ and an index $1 \leq t \leq \ell - 1$. Apply (3) with $p = x^{t-1}$ and $r = x^{t+1}$, and let

$$N = \alpha_t + \alpha_{t+1} = u(r) - u(p).$$

By Lemma 3.1 the admissible replacements for x^t form the box $\prod_i [L_i, U_i]$, and by (2) the resulting value of α_t is $u(x^t) - u(p)$, which ranges over *every* integer in

$$[\mu_-, \mu_+], \quad \mu_- = \sum_{i=1}^m L_i - u(p), \quad \mu_+ = \sum_{i=1}^m U_i - u(p),$$

since u is a sum of independently varying integer coordinates. Lemma 3.2 says precisely

$$\mu_- + \mu_+ = N. \quad (4)$$

Corollary 3.4 (Cylindric Bender–Knuth; Postnikov’s symmetry theorem). *The map ι_t which replaces x_i^t by $L_i + U_i - x_i^t$ for all i , and leaves all other rows unchanged, is an involution of $\mathcal{T}_\ell(\lambda/\mu)$ which interchanges α_t and α_{t+1} and fixes all other α_s . Consequently $s_{\lambda/\mu}^c$ is a symmetric function, and $W_\ell(\lambda/\mu)$ is stable under S_ℓ .*

Proof. Reflection in a box is an involution of the box, so by Lemma 3.1 the result is again a chain (the rows x^s , $s \neq t$, are untouched, and only the steps t and $t+1$ involve x^t). Its weight in position t is

$$\sum_{i=1}^m (L_i + U_i - x_i^t) - u(p) \stackrel{3.2}{=} u(p) + u(r) - u(x^t) - u(p) = u(r) - u(x^t) = \alpha_{t+1},$$

and since $\alpha_t + \alpha_{t+1}$ is unchanged, position $t+1$ receives α_t . The involutions $\iota_1, \dots, \iota_{\ell-1}$ realise the adjacent transpositions of S_ℓ on weights, which generate S_ℓ . $\square$

Corollary 3.5 (The exchange move). *Let $\alpha \in W_\ell(\lambda/\mu)$ and let $1 \leq t \leq \ell-1$ satisfy $\alpha_t > \alpha_{t+1}$. Then $\alpha - e_t + e_{t+1} \in W_\ell(\lambda/\mu)$. Moreover the move is realised by sliding a single bead of the row x^t one site to the left.*

Proof. Let T be a chain with weight α and form the box of Lemma 3.1 at position t . If $x_i^t = L_i$ for every i then $\alpha_t = \mu_-$, and (4) gives $\mu_- \leq N/2$, i.e. $\alpha_t \leq N - \alpha_t = \alpha_{t+1}$, contrary to hypothesis. Hence $x_j^t > L_j$ for some $j \in \{1, \dots, m\}$. Replacing x_j^t by $x_j^t - 1$ (and, by periodicity, x_{j+km}^t by $x_{j+km}^t - 1$ for all k) stays inside the box, hence gives a chain, by Lemma 3.1; its weight is $\alpha - e_t + e_{t+1}$. $\square$

4 The weight set is a dominance ideal

For $\alpha \in \mathbb{N}^\ell$ let $\text{sort}(\alpha)$ be the partition obtained by sorting its positive entries decreasingly, and let $P(J) = \{\text{sort}(\alpha) : \alpha \in J\}$ for $J \subseteq \mathbb{N}^\ell$. Recall $\sigma \leq \nu$ (dominance) means $\sigma_1 + \dots + \sigma_r \leq \nu_1 + \dots + \nu_r$ for all r , for partitions of the same size.

Lemma 4.1 (Robin Hood step). *Let $\nu \triangleright \sigma$ be partitions of d . Then there are indices $a < b$ with $\nu_a \geq \nu_b + 2$ such that $\tau := \nu - e_a + e_b$ is again a partition and $\nu \triangleright \tau \supseteq \sigma$.*

Proof. Let i be minimal with $\nu_i > \sigma_i$ (it exists and $\nu_c = \sigma_c$ for $c < i$). Let j be minimal with $j > i$ and $\nu_j < \sigma_j$; such j exists, for otherwise $\nu_c \geq \sigma_c$ for all $c \geq i$, which together with $\nu_i > \sigma_i$ contradicts $|\nu| = |\sigma|$. Then $\nu_i > \sigma_i \geq \sigma_j > \nu_j$, so $\nu_i \geq \nu_j + 2$.

Writing $N_r = \nu_1 + \dots + \nu_r$ and $S_r = \sigma_1 + \dots + \sigma_r$, we claim $S_r < N_r$ for $i \leq r < j$. For $r = i$: $S_i = S_{i-1} + \sigma_i = N_{i-1} + \sigma_i < N_{i-1} + \nu_i = N_i$. For $i < r < j$: by minimality of j we have $\sigma_c \leq \nu_c$ for $i < c < j$, so $S_r \leq S_i + \sum_{c=i+1}^r \nu_c \leq (N_i - 1) + \sum_{c=i+1}^r \nu_c = N_r - 1$.

Now set $a = \max\{c : i \leq c < j, \nu_c = \nu_i\}$ and $b = \min\{c : i < c \leq j, \nu_c = \nu_j\}$. Since $\nu_i > \nu_j$ these lie in different constancy blocks of ν , so $a < b$, and $\nu_a = \nu_i \geq \nu_j + 2 = \nu_b + 2$. By the choice of a and b we have $\nu_{a+1} < \nu_a$ and $\nu_{b-1} > \nu_b$, so $\tau = \nu - e_a + e_b$ is weakly decreasing, i.e. a partition (if $b = a + 1$, then $\nu_a - 1 \geq \nu_b + 1$ and this is again clear). Its partial sums are $T_r = N_r - [a \leq r < b]$, and $[a, b] \subseteq [i, j)$, so $S_r \leq T_r$ for all r by the claim. Hence $\sigma \trianglelefteq \tau \triangleleft \nu$. $\square$

Proposition 4.2. $P(W_\ell(\lambda/\mu))$ is an order ideal of the dominance order on partitions of d with at most ℓ parts.

Proof. Let $\nu \in P(W_\ell)$ and $\sigma \trianglelefteq \nu$; we show $\sigma \in P(W_\ell)$. By Lemma 4.1 and induction on the (finite) number of steps, it suffices to treat $\sigma = \tau = \nu - e_a + e_b$ with $a < b$ and $\nu_a \geq \nu_b + 2$.

By Corollary 3.4, W_ℓ is S_ℓ -stable, so it contains a vector β whose entries are those of ν (padded with zeros) arranged so that $\beta_t = \nu_a$ and $\beta_{t+1} = \nu_b$ for some t , the remaining entries being the remaining parts of ν in any order. Since $\beta_t = \nu_a > \nu_b = \beta_{t+1}$, Corollary 3.5 gives $\beta - e_t + e_{t+1} \in W_\ell$, and $\text{sort}(\beta - e_t + e_{t+1}) = \tau$. $\square$

5 The greedy chain and the dominance maximum

Definition 5.1. The *greedy chain* for λ/μ is defined by $g^0 = \mu$ and

$$g_i^t = \min(g_{i+1}^{t-1} - 1, \lambda_i), \quad i \in \mathbb{Z}, t \geq 1. \quad (5)$$

Lemma 5.2. Assume $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$. Then:

1. each g^t is a cylindric shape with $g^{t-1} \ll g^t$ and $g^t \subseteq \lambda$;
2. $g^\ell = \lambda$, so $G := (g^0, \dots, g^\ell) \in \mathcal{T}_\ell(\lambda/\mu)$;
3. for every $T = (x^0, \dots, x^\ell) \in \mathcal{T}_\ell(\lambda/\mu)$ and every t one has $x_i^t \leq g_i^t$ for all i ; in particular $u(x^t) \leq u(g^t)$.

Proof. (1) g^t is n -periodic and strictly increasing, being a coordinatewise minimum of two strictly increasing n -periodic sequences. Assuming inductively $g^{t-1} \subseteq \lambda$ we get $g_i^{t-1} \leq g_{i+1}^{t-1} - 1$ and $g_i^{t-1} \leq \lambda_i$, so $g_i^{t-1} \leq g_i^t$; and $g_i^t \leq g_{i+1}^{t-1} - 1 < g_{i+1}^{t-1}$, so $g^{t-1} \ll g^t$ by (1). Also $g_i^t \leq \lambda_i$.

(3) Let $T \in \mathcal{T}_\ell$. Then $x^t \subseteq x^\ell = \lambda$ for all t . We induct: $x^0 = g^0$; and if $x_i^{t-1} \leq g_i^{t-1}$ for all i , then $x^t \ll$ -condition gives $x_i^t \leq x_{i+1}^{t-1} - 1 \leq g_{i+1}^{t-1} - 1$, while $x_i^t \leq \lambda_i$; hence $x_i^t \leq g_i^t$.

(2) By (3) applied to any $T \in \mathcal{T}_\ell$ (nonempty by hypothesis), $\lambda_i = x_i^\ell \leq g_i^\ell \leq \lambda_i$. $\square$

Definition 5.3. Let $\gamma_t = u(g^t) - u(g^{t-1})$ be the weight of the greedy chain and $\hat{\lambda} = \hat{\lambda}(\lambda/\mu, \ell) = \text{sort}(\gamma)$.

Proposition 5.4. $\hat{\lambda} \in P(W_\ell)$ and $\nu \trianglelefteq \hat{\lambda}$ for every $\nu \in P(W_\ell)$.

Proof. $\gamma \in W_\ell$ by Lemma 5.2(2), and $\hat{\lambda} = \text{sort}(\gamma) \in W_\ell$ by Corollary 3.4; so $\hat{\lambda} \in P(W_\ell)$.

Let $\nu \in P(W_\ell)$ and $r \geq 1$. Since W_ℓ is S_ℓ -stable, ν itself (padded with zeros) lies in W_ℓ , say $\nu = \alpha(T)$ for some $T = (x^0, \dots, x^\ell)$. Telescoping (2),

$$\nu_1 + \dots + \nu_r = u(x^r) - u(x^0) \stackrel{5.2(3)}{\leq} u(g^r) - u(g^0) = \gamma_1 + \dots + \gamma_r \leq \hat{\lambda}_1 + \dots + \hat{\lambda}_r,$$

the last inequality because $\hat{\lambda}_1 + \dots + \hat{\lambda}_r$ is the maximum of $\gamma_{t_1} + \dots + \gamma_{t_r}$ over all r -subsets. Hence $\nu \trianglelefteq \hat{\lambda}$. $\square$

6 From a dominance ideal with a maximum to M-convexity

This section is self-contained polymatroid theory; we include it so that the proof cites nothing.

Lemma 6.1. *Let $\rho : 2^{[\ell]} \rightarrow \mathbb{Z}$ be submodular and let $J = \{\alpha \in \mathbb{N}^\ell : \alpha([\ell]) = \rho([\ell]), \alpha(S) \leq \rho(S) \forall S\}$, where $\alpha(S) = \sum_{i \in S} \alpha_i$. For $\alpha \in J$ the family $\mathcal{F}_\alpha = \{S : \alpha(S) = \rho(S)\}$ is closed under union and intersection.*

Proof. For $S, T \in \mathcal{F}_\alpha$, $\rho(S) + \rho(T) = \alpha(S) + \alpha(T) = \alpha(S \cup T) + \alpha(S \cap T) \leq \rho(S \cup T) + \rho(S \cap T) \leq \rho(S) + \rho(T)$, forcing equality throughout. $\square$

Proposition 6.2. *Let $\hat{\lambda}$ be a partition of d with at most ℓ parts and set $\Lambda_r = \hat{\lambda}_1 + \dots + \hat{\lambda}_r$ for $0 \leq r \leq \ell$. Then*

$$J = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \preceq \hat{\lambda}\} = \{\alpha \in \mathbb{N}^\ell : \alpha([\ell]) = d, \alpha(S) \leq \Lambda_{|S|} \forall S \subseteq [\ell]\}$$

is M-convex.

Proof. The displayed equality holds because $\max_{|S|=r} \alpha(S)$ is the sum of the r largest entries of α . Put $\rho(S) = \Lambda_{|S|}$. As $\hat{\lambda}$ is weakly decreasing, $r \mapsto \Lambda_r$ is concave; and for any S, T we have $|S \cup T| + |S \cap T| = |S| + |T|$ with $|S \cap T| \leq \min(|S|, |T|) \leq \max(|S|, |T|) \leq |S \cup T|$, so concavity gives $\Lambda_{|S \cup T|} + \Lambda_{|S \cap T|} \leq \Lambda_{|S|} + \Lambda_{|T|}$: ρ is submodular.

Let $\alpha, \beta \in J$ and i with $\alpha_i > \beta_i$. Note $\alpha - e_i + e_j \in \mathbb{N}^\ell$ for any j , since $\alpha_i \geq 1$. For $S \subseteq [\ell]$, $(\alpha - e_i + e_j)(S) = \alpha(S) - [i \in S] + [j \in S]$ exceeds $\rho(S)$ only if $j \in S$, $i \notin S$ and $S \in \mathcal{F}_\alpha$. Hence

$$\alpha - e_i + e_j \notin J \iff \exists S \in \mathcal{F}_\alpha \text{ with } j \in S, i \notin S.$$

Let $D = \{j : \alpha_j < \beta_j\}$; $D \neq \emptyset$ because $\alpha([\ell]) = \beta([\ell])$ and $\alpha_i > \beta_i$, and $i \notin D$. Suppose, for contradiction, that $\alpha - e_i + e_j \notin J$ for every $j \in D$, and pick $S_j \in \mathcal{F}_\alpha$ with $j \in S_j$, $i \notin S_j$. By Lemma 6.1, $S^* = \bigcup_{j \in D} S_j \in \mathcal{F}_\alpha$, and $D \subseteq S^*$, $i \notin S^*$. Then

$$\beta(S^*) - \alpha(S^*) = \sum_{j \in S^*} (\beta_j - \alpha_j) \geq \sum_{j \in D} (\beta_j - \alpha_j) > 0,$$

the first inequality because every term with $j \in S^* \setminus D$ is ≤ 0 , and the second because $D \neq \emptyset$. Hence $\beta(S^*) > \alpha(S^*) = \rho(S^*)$, contradicting $\beta \in J$. $\square$

7 The main theorem

Theorem 7.1. *Let λ/μ be a cylindric skew shape in $C^{n,m}$, let $\ell \geq 1$, and suppose $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$. Let $\hat{\lambda} = \hat{\lambda}(\lambda/\mu, \ell)$ be the sorted weight of the greedy chain of Definition 5.1. Then*

$$\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = W_\ell(\lambda/\mu) = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \preceq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell,$$

and this set is M-convex. Consequently $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ is M-convex, has saturated Newton polytope, and $\text{Newton}(s_{\lambda/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$.

Proof. By Proposition 5.4, every $\nu \in P(W_\ell)$ satisfies $\nu \preceq \hat{\lambda}$, and $\hat{\lambda} \in P(W_\ell)$. By Proposition 4.2, $P(W_\ell)$ is a dominance ideal, so it contains every $\sigma \preceq \hat{\lambda}$. Hence $P(W_\ell) = \{\sigma : \sigma \preceq \hat{\lambda}\}$, and since W_ℓ is S_ℓ -stable (Corollary 3.4), $W_\ell = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \preceq \hat{\lambda}\}$. That this equals $\mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell$ is the (elementary) statement that the lattice points of the $\hat{\lambda}$ -permutahedron are exactly the α with $\text{sort}(\alpha) \preceq \hat{\lambda}$; it is also not needed for M-convexity, which follows from Proposition 6.2. SNP and the identification of the Newton polytope follow since $\mathcal{P}_{\hat{\lambda}} = \text{conv}(W_\ell)$. $\square$

Remark 7.2. If $\mathcal{T}_\ell(\lambda/\mu) = \emptyset$ then $s_{\lambda/\mu}^c(x_1, \dots, x_\ell) = 0$ and there is nothing to prove. The greedy chain also computes the minimal ℓ for which $\mathcal{T}_\ell \neq \emptyset$: it is the least t with $g^t = \lambda$.

8 What the proof consumes

The success criterion attached to WZZ Problem 5.2 is that the argument must not route through [2, Cor. 8.5]. We therefore list every external input.

- **Lam, Cor. 8.5** (affine Stanley $\Rightarrow$ dual k -Schur positivity): *not used*.
- **Lee 2019, Cor. 5** (cylindric skew Schur = affine Stanley for a 321-avoiding affine permutation): *not used*.
- **WZZ Theorem 3.6** (k -Kostka positivity) and **Theorem 4.4**: *not used*.
- **Rado’s theorem** ($\mathcal{P}_\mu \subseteq \mathcal{P}_\lambda \iff \mu \trianglelefteq \lambda$): *not used*. Proposition 6.2 proves M-convexity directly from the exchange axiom.
- **Postnikov’s symmetry theorem** for cylindric skew Schur functions: *not used* — it is reproved as Corollary 3.4.
- **Murota’s theory of M-convex sets**: only the *definition* is used.
- **Classical facts about partitions**: that λ/μ is a horizontal strip iff $\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \dots$, and the β -set encoding of a partition (Macdonald I.(1.7)). These are used once, inside Proposition 2.2, to translate the lattice-path definition into the bead model.

The proof is therefore combinatorial and based on the tableau interpretation, as Problem 5.2 requests.

Remark 8.1 (What the answer looks like as a tableau statement). Unwinding Corollary 3.5 through Proposition 2.2: given a semistandard cylindric skew tableau in which the letter t occurs more often than the letter $t+1$ (per period), one may change a single box from t to $t+1$. The box to change is located by the bead picture: the row λ^t of the chain is not at the left end of its box $\prod_i [L_i, U_i]$, and retracting one bead of λ^t by one site is exactly the required box change. The cylindric Bender–Knuth involution (Corollary 3.4) is the full reflection of that box.

9 Computational verification

All computations were run in Python over the bead model of §2; the code is at [projects/scratch/q191/code/cyl](https://github.com/q191/code/cyl).

Anchor (is the model the right object?). For $n = 20$, $m = 2$, $\mu = (0, 2)$, $\lambda = (3, 7)$ — a cylinder wide enough that the shape does not wrap — the bead model’s weight generating function agrees in all 13 coefficients with an independently coded ordinary skew Kostka computation for the corresponding skew partition shape $(8, 5)/(3, 2)$.

Positive control (WZZ Example 4.2). WZZ record, for $w = s_2 s_1 s_0 s_2 \in \tilde{S}_3$, $\tilde{F}_w = m_{1111} + m_{211} + m_{22} = s_{22} - s_{1111}$. A search over $\mathcal{C}^{3,m}$ ($m = 1, 2$) with $d = 4$ finds exactly one realisation, $m = 1$, $\mu = (0)$, $\lambda = (4)$, and its monomial expansion is $m_{1111} + m_{211} + m_{22}$; converting to the Schur basis gives $s_{22} - s_{1111}$, reproducing the published value including the negative coefficient. This control is untuned (published by other authors for exactly this purpose), signed, and has M-convex support — so it also certifies that the model is *not* computing ordinary skew Schur functions in disguise.

Lemma 3.1 (decoupling). For all $2 \leq n \leq 5$, $1 \leq m < n$, all p with $p_1 = 0$, and all $r \supseteq p$ within a window of 5: the set of admissible middle shapes coincides with $\prod_i [L_i, U_i]$ in all 380 feasible instances (290 middle shapes in total). No failures.

Lemma 3.2 (palindrome identity). 470 feasible (p, r) pairs over $2 \leq n \leq 6$: $\mu_- + \mu_+ = N$ in all 470. (This computation corrected an algebra slip in my first derivation, which dropped the wrap-around term and gave $N - n$.)

Lemma 5.2 and Proposition 5.4 (greedy maximum). 462 instances over $2 \leq n \leq 5$, $d \leq 7$, $\ell \leq 5$: in every one the greedy chain reaches λ in ℓ steps and $\text{sort}(\gamma)$ is a member of $P(W_\ell)$ dominating every element of $P(W_\ell)$. No failures.

Theorem 7.1. 1932 instances over $2 \leq n \leq 6$, $d \leq 7$: $P(W_\ell)$ is a dominance ideal with a unique maximal element in every one. Separately, 409 instances were checked against the raw definition of M-convexity by exhaustive search over all pairs $(\alpha, \beta) \in W_\ell^2$ and all i ; all pass, and in all 409 the support is S_ℓ -stable, confirming Corollary 3.4. Finally, 339 instances were checked for symmetry of the *coefficients* (not merely the support); all pass.

10 Gaps

I record the following honestly.

1. **Proposition 2.2 is proved by local reduction.** The argument that the three conditions (containment, box count, horizontal strip) are local, and therefore may be checked in a bounded window where a cylindric shape is an ordinary partition, is correct but is presented compactly. A referee would want the window argument spelled out with the explicit correspondence between the n -periodic bead set and the β -set of the partition truncated to the window. I regard this as routine rather than as a gap in the mathematics, but it is the only step I have not written at full length, and it is the step at which a convention error would produce a correct theorem about the wrong object. It is for that reason that the anchor and the positive control of §9 were run first.
2. $\mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell = \{\alpha : \text{sort}(\alpha) \preceq \hat{\lambda}\}$ is asserted without proof in Theorem 7.1. It is Rado's theorem, and it is used only for the cosmetic identification of the Newton polytope; M-convexity, SNP and the description of the support do not depend on it.
3. **Dependence of $\hat{\lambda}$ on ℓ .** $\hat{\lambda}(\lambda/\mu, \ell)$ grows with ℓ until ℓ reaches the minimal chain length, after which it is constant. I have not written out the (easy) monotonicity statement.

References

- [1] Wang, Zhang, Zhang, *Newton polytopes and M-convexity of dual k -Schur polynomials*, arXiv:2401.14632. (Read at LaTeX source; M-convexity definition at line 122, Theorem 3.6, Theorem 4.4, Corollary 4.9, Problems 5.1–5.2 at lines 938 and 945.)
- [2] T. Lam, *Schubert polynomials for the affine Grassmannian*, J. Amer. Math. Soc. 21 (2008). (Cited only to say it is *not* used.)
- [3] S. J. Lee, *Cylindric skew Schur functions and affine Stanley symmetric functions*. (Cited only to say it is *not* used.)
- [4] L. Naprienko, *Free fermionic Schur functions*, arXiv:2301.12110. (Read at LaTeX source; Lemma 4.4 `lem:ribbonmaya` at line 1528, bead/hole conventions at eq. (4.1).)
- [5] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., I.(1.7).